

Geonwoo Kim

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“There is no failure, only growth.”

Education

Inha University, Department of Artificial Intelligence

Incheon, S.Korea

B.S. IN ARTIFICIAL INTELLIGENCE, JUNIOR

Mar. 2021 – Present

- GPA: **4.16 / 4.5** (Rank: 8 / 59)

Research Interests

Generative Models, Multimodal Learning, Computer Vision

Research Experience

DAVIAN Lab | Kim Jaechul Graduate School of AI, KAIST

Seoul, S.Korea

UNDERGRADUATE RESEARCH INTERN

Dec. 2025 – Present

- Conducting research on **video generation**.
- Advised by Professor Jaegul Choo.

Multimodal AI Lab | Inha University

Incheon, S.Korea

UNDERGRADUATE RESEARCHER

Dec. 2024 – Present

- Conducting research on **subject-driven image editing** using **multimodal inputs** (image, mask, and text) and **in-context learning**.
- Participated in **weekly seminars** and delivered **monthly research presentations**.

Project Experience

CoverUs: AI-based Comment Management and Creator Insight Solution

Incheon, S.Korea

TEAM LEADER, PRESENTER

Jun. 2025 – Sep. 2025

- Designed an **AI comment management solution** integrating **MCP-based data collection**, **RAG retrieval**, and tone-adaptive response generation for creators.
- Implemented a two-stage learning pipeline combining few-shot GPT-5 inference and LoRA fine-tuning for **personalized responses**.
- Developed **similarity-based comment retrieval** and sentiment clustering modules for creator–fan insight extraction.
- Built full-stack MVP (React + FastAPI) and received the **Excellence Award** for innovation and business feasibility.

COSFLOW: AI Automation Solution for Cosmetic OEM Companies

Incheon, S.Korea

TEAM MEMBER

Jun. 2025 – Sep. 2025

- Developed an **AI Agent solution** streamlining cosmetic ingredient compliance workflows for OEM clients.
- Built an **LLM-integrated pipeline** that standardizes ingredient names, queries multi-country databases, and generates structured compliance reports (PDF).
- Implemented **vector database-based ingredient similarity search** and GPT-4.1–driven report generation, including ingredient cards and dashboards.

Text-Guided Image Colorization (Inha AI Challenge)

Incheon, S.Korea

TEAM LEADER

Jun. 2025 – Aug. 2025

- Led an AI competition project focused on text-guided colorization of grayscale images.
- Built a **multimodal diffusion-based framework** combining a **DiT backbone** with dual-decoder colorization.
- Designed ensemble pipelines integrating text embeddings and image reconstruction networks to improve HSV and CLIP similarity.
- Achieved the **BK21 Director’s Award (Excellence)** with a final score of **0.6178**.

AI-Generated vs. Human-Written Text Classification Challenge

Seoul, S.Korea

TEAM LEADER

Jun. 2025 – Jul. 2025

- Developed an AI model distinguishing human-written and LLM-generated Korean texts, despite full-text–level labeling.
- Improved paragraph-level discrimination via **LLM-based data augmentation** and **ensemble modeling**.
- Used RoBERTa-Large, BERT-base, KoBigBird embeddings with 3:1 undersampling to address imbalance.
- Achieved **ROC-AUC 0.906 (Top 11%)** among 251 teams.

Enterprise-tailored AI Research & Report Generation Agent

Incheon, S.Korea

TEAM MEMBER

Mar. 2025 – Jun. 2025

- Developed an **AI Agent system integrating RAG retrieval**, vector database search, and template-based report automation.
- Built a **LangChain-powered CSV Analysis Agent** supporting data queries, visualization, and automated summaries in Streamlit.
- Designed a custom GPT-5 pipeline for domain-adaptive generation and reliability evaluation of external sources.
- Secured **₩80M in government funding** through the **Korean Pre-Startup Package and Lab-based Startup Leading University Program**.

OTHERS

- **MNIST Dataset Image Classification and K-Means Clustering** (Using TensorFlow, Scikit-learn)
- **Image Classification via Transfer Learning** (Using EfficientNetV2, ResNet)
- **Sequence-to-Sequence Neural Machine Translation** (Using RNN Encoder–Decoder, Attention Mechanism)
- **IMDB Sentiment Analysis using RNN, LSTM, and GRU** (Using PyTorch, TorchText)
- **Fine-tuning IMDB Sentiment Classifier using LoRA** (Using HuggingFace Transformers, PEFT)
- **Feature Mapping and Kernelized K-Means for Nonlinear Data Classification and Clustering** (Using Scikit-learn, NumPy)

Awards & Honors

AWARDS

2025.09	Director's Award (Excellence, Final Round) , Incheon University-wide Startup Competition	Incheon, S.Korea
2025.08	BK21 Director's Award (Excellence) , AI Competition on Text-Guided Image Colorization	Inha Univ.
2025.06	Mayor's Award (Outstanding Idea, Preliminary Round) , Incheon University-wide Startup Competition	Incheon, S.Korea
2025.05	Grand Prize (President's Award) , Inter-university IT Hackathon(MOONG)	Inha Univ.
2025.03	Grand Prize , Startup Competition – AI Agent-based Enterprise Search Engine SaaS	Inha Univ.
2024.11	1st Place , Google Developer Groups IT Hackathon Ideathon and Demo Day	Inha Univ.
2022.09	1st Place (Dean's Award) , Academic Excellence	Inha Univ.

HONORS

2021	Academic Excellence Scholarship , Awarded to students with outstanding academic achievement	Inha Univ.
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Extracurricular Activity

AI Society: SINSA

Incheon, S.Korea

VICE PRESIDENT

Jun. 2025 – Present

- Managing a **130-member university AI society** under faculty supervision, overseeing programs in Python, Data Analysis, Deep Learning, and Paper Study.
- Organized internal **AI competitions** and structured **study curricula**, contributing to multiple national awards in SW-centered University AI competitions.

RAG/Agent Study

Incheon, S.Korea

STUDY LEADER

Aug. 2025 – Present

- Led a study group focused on **Retrieval-Augmented Generation (RAG)** and **LLM-based Agents** using LangChain, LangGraph, and MCP frameworks.
- Conducted both theoretical exploration and hands-on implementation of **multimodal RAG/Agent systems**, followed by discussions on their research and industrial applications.

Bishop–Deep Learning Study

Incheon, S.Korea

STUDY MEMBER

Mar. 2025 – Present

- Participating in a year-long study based on Bishop's Deep Learning: Foundations and Concepts, covering theoretical and practical aspects.
- Deliver weekly presentations on **CNNs, Diffusions, and Transformers** through peer teaching and collaborative discussions.

Start-up AI Agent Lab: UPFLOW

Incheon, S.Korea

AI RESEARCHER

Mar. 2025 – Sep. 2025

- Led **AI Agent startup** recognized by the Pre-Startup Package and Lab-based Startup Program (**₩80M government grant**).
- Developed enterprise-level **AI Agent** and **MCP-based automation systems** for consulting and cosmetic OEM clients.
- Acquired **end-to-end experience** from model development to commercialization through data-driven, problem-solving projects.

CS231n Study (Stanford Course)

Incheon, S.Korea

STUDY MEMBER

Mar. 2025 – Jun. 2025

- Participated in a weekly discussion-based study group on **Stanford CS231n: Convolutional Neural Networks for Visual Recognition**.
- Presented CNN architectures and applications in **computer vision**, facilitating discussions on core lecture topics.

OTHERS

- **Participated in ISW SURF (Incheon Startup Week)** – Operated exhibition booth and delivered IR pitching session at Songdo Convensia.
- **Completed Micro-major in Business Insight** – Fulfilled required coursework (Marketing Insight, Principles of Brand Creation, Leadership in Persuasion)
- **Completed 4th Smart Semiconductor Academy** – Studied AI semiconductor fundamentals, large language models, accelerator design, and next-generation memory.
- **Research Intern, Incheon Living Lab Project** – Conducted joint research with AI companies and public officials in Michuhol-gu.
- **Attended Academic Events** – Participated in AI Graduate Symposium, AI Expo Korea, and AI Convergence Center International Symposium, etc.

Teaching

Environmental Data Analysis Mentoring (Incheon Metropolitan Office of Education)

Incheon, S.Korea

MENTOR

May. 2025 – Jun. 2025

- Conducted **AI-based environmental data analysis mentoring** for middle and high school students under the Incheon Metropolitan Office of Education.
- Utilized NLP techniques such as **KoNLPy, stopwords removal, and Kkma morphological analysis** to identify and analyze carbon-neutrality issues.
- Implemented a recycling-status **dashboard and an object detection model** for recyclable item classification.

Digital Green Workshop (Inha University Digital Innovation Center)

Incheon, S.Korea

MENTOR

Oct. 2024

- Mentored high school students in an environmental data analysis workshop organized by Inha University and the Incheon Metropolitan Office of Education.
- Guided participants to define environmental problems through **AI-based data analysis** and propose **blockchain incentive systems** as practical solutions.

EBS 1318 On-Coaching Program

Siheung, S.Korea

MENTOR

Jul. 2024 – Present

- Provided one-on-one mathematics and science tutoring for middle and high school students.
- Focused on fundamental topics such as **equations, functions, and set theory**, emphasizing conceptual understanding.

Hwasung Mathematics Academy

Yongin, S.Korea

TEACHING ASSISTANT

Mar. 2021 – Feb. 2022

- Assisted with curriculum design and personalized learning plans based on student needs.
- Taught mathematics including **probability, statistics, and calculus** for college entrance preparation.
- Communicated with students and parents to coordinate academic goals and progress.

Skills

Programming	Python, C++
AI Framework	PyTorch, TensorFlow
RAG & Agents	LangChain, Streamlit, MCP
Tools	Linux, SSH, GitHub
Libraries	NumPy, Pandas, Matplotlib, Scikit-learn

Others

Certification

ADSP (MAR. 2024)